

LECTURE 1: SEGMENTATION

FROM U-NET TO NNU-NET — ONE LABEL PER PIXEL

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SEGMENTATION IS A DENSE CLASSIFICATION TASK

- Classification: **one** label per image.
- Segmentation: **one** label per pixel.

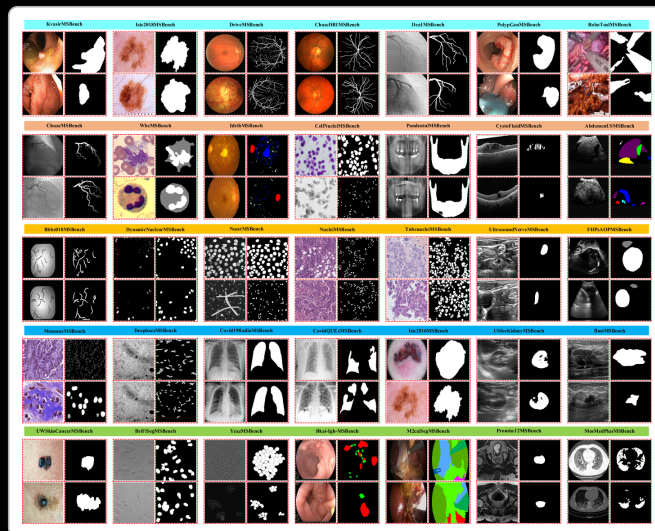
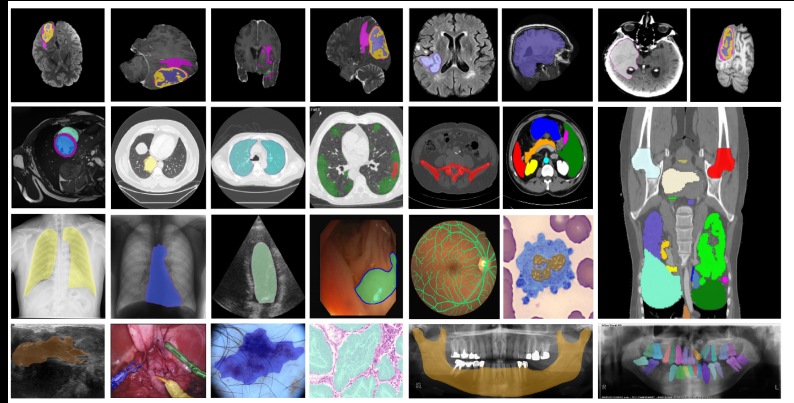


Fig. 1, Kuş & Aydın, *MedSegBench*, Scientific Data 11, 1283 (2024), CC BY-NC-ND 4.0.

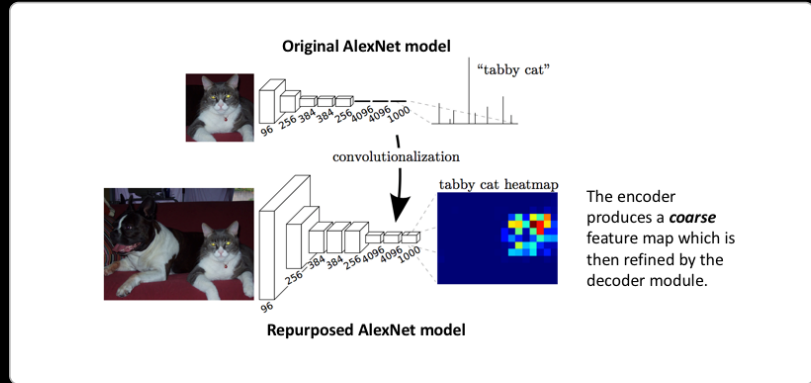
WHY CLINICIANS WANT MASKS, NOT LABELS



Medical image segmentation is used for **diagnosis**, **treatment planning**, and **patient care** — because a label cannot be measured, and a mask can.

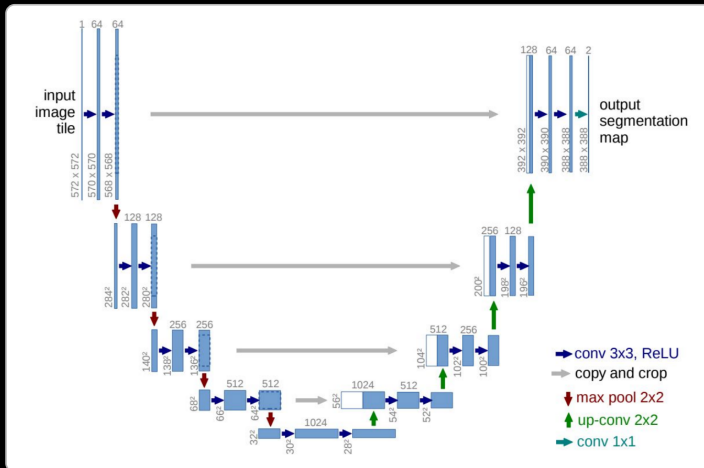
The U-Net

THE RESOLUTION PROBLEM



Segmentation needs a **full-resolution** output.

THE U-NET



Contracting path

Two 3×3 **unpadded convolutions** + ReLU, then 2×2 **max pooling**, stride 2. Each step **doubles the feature channels**.

Expansive path

Upsampling, then a 2×2 **up-convolution** that **halves the feature channels**, and two 3×3 convolutions + ReLU.

Copy and crop

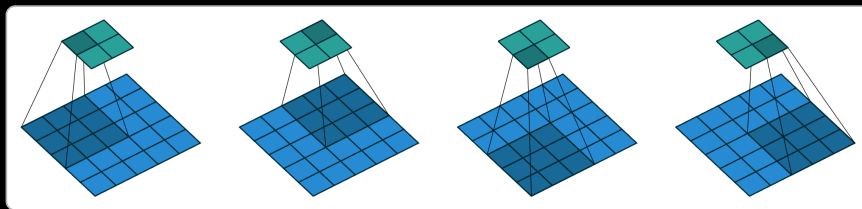
Concatenation with the **correspondingly cropped** feature map from the contracting path — every unpadded convolution loses border pixels.

Terminology after Ronneberger, Fischer & Brox, *U-Net: Convolutional Networks for Biomedical Image Segmentation*, MICCAI 2015

(arXiv:1505.04597), §2.

CONVOLUTION (2D)

A 3×3 kernel over a 5×5 input, stride 2 — output 2×2 .



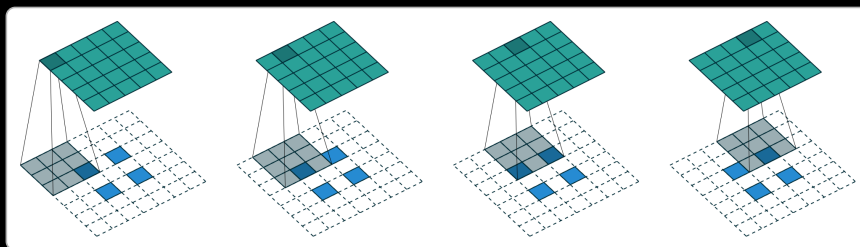
$$y[r, c] = \sum_{m=0}^{k-1} \sum_{n=0}^{k-1} x[sr + m, sc + n] w[m, n]$$

$$\text{output size } o = \left\lfloor \frac{i + 2p - k}{s} \right\rfloor + 1 = \left\lfloor \frac{5 + 0 - 3}{2} \right\rfloor + 1 = 2$$

Figure 2.5 from Dumoulin & Visin, *A guide to convolution arithmetic for deep learning* (arXiv:1603.07285). Input blue, output green.

TRANPOSED CONVOLUTION

The transpose of that convolution: 2×2 in, 5×5 out.



$$y[sr + m, sc + n] += x[r, c] w[m, n]$$

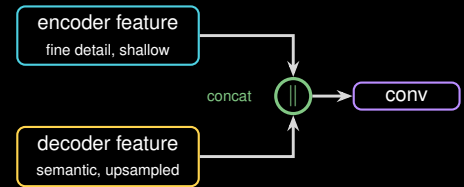
$$\text{output size } o = s(i - 1) + k - 2p = 2(2 - 1) + 3 - 0 = 5$$

$$y = C^T x \quad \text{— hence } \textit{transposed}.$$

Figure 4.5 from Dumoulin & Visin (arXiv:1603.07285) — the transpose of Figure 2.5 on the previous slide.

SKIP CONNECTIONS: WHERE THE BOUNDARIES COME FROM

- ▶ Upsample the coarse decoder feature, then **concatenate** the matching encoder feature map.
- ▶ The encoder map still holds the *fine spatial detail* that pooling destroyed.
- ▶ Result: semantics from the deep path, **precise boundaries** from the shallow one.
- ▶ Bonus: a short gradient path to the early layers, exactly as in ResNet.



Concatenate, not add — the decoder learns how to weigh the two sources.

How do we know it worked?

IoU (JACCARD INDEX)

Intersection over union

$$\text{IoU} = \frac{TP}{TP + FP + FN}$$

Pixels in *both* masks, over pixels in *either*. Range [0, 1], higher is better.

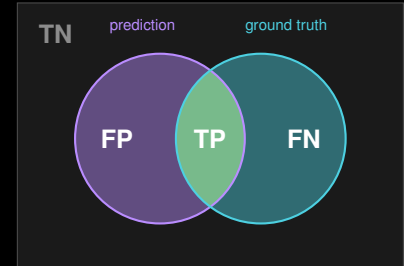
Work one through

Prediction 100 px, ground truth 120 px, overlap 80 px.

$TP=80$, $FP=20$, $FN=40$.

$\text{IoU} = 80/140 = \mathbf{0.57}$ $\text{Dice} = 160/220 = \mathbf{0.73}$

Compute **per image**, then average — not pooled over the test set, which lets big easy cases hide small hard ones.



TN never appears — which is exactly why IoU survives class imbalance and accuracy does not.

DICE COEFFICIENT

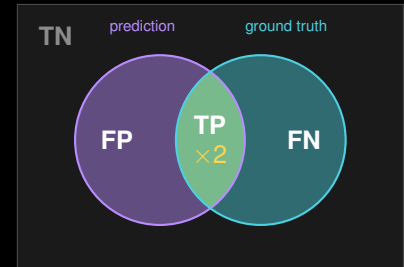
The metric medical imaging reports

$$\text{DSC} = \frac{2 TP}{2 TP + FP + FN}$$

Twice the intersection, over the sum of the two mask sizes — and again, no *TN*.

Also known as

The **F1 score** computed over pixels: precision = $TP/(TP+FP)$ and recall = $TP/(TP+FN)$, harmonic mean. Also called DSC, the Dice similarity coefficient.



TP is counted **twice** in the numerator — that is the whole difference from IoU.

DICE VS. IOU: NEVER ARGUE ABOUT WHICH IS BETTER

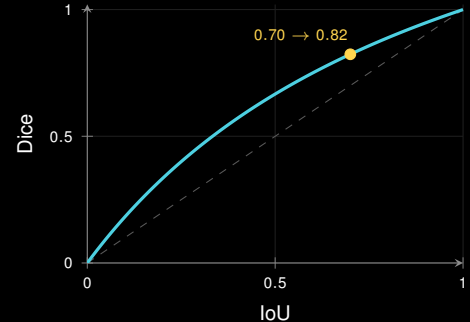
They are the same ranking

$$\text{Dice} = \frac{2 \text{IoU}}{1 + \text{IoU}} \quad \text{IoU} = \frac{\text{Dice}}{2 - \text{Dice}}$$

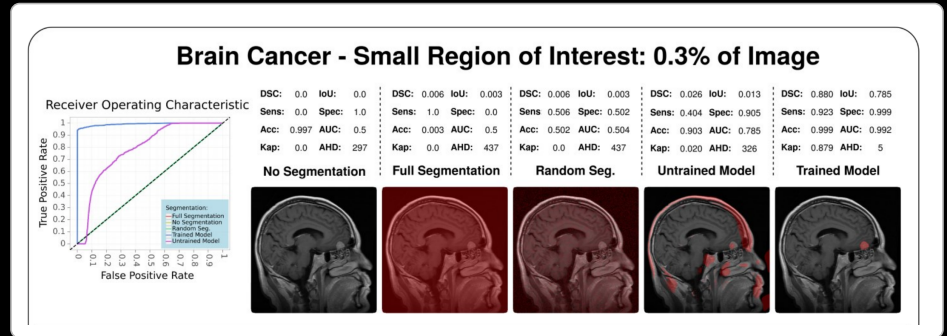
A strictly increasing map. If model A beats B on Dice, it beats B on IoU.

But the numbers differ

Dice is **never smaller** than IoU, and strictly larger whenever the match is imperfect. IoU 0.70 is Dice 0.82. Check which one a paper quotes before you compare against it.



ACCURACY IS HIGHLY DISCOURAGED

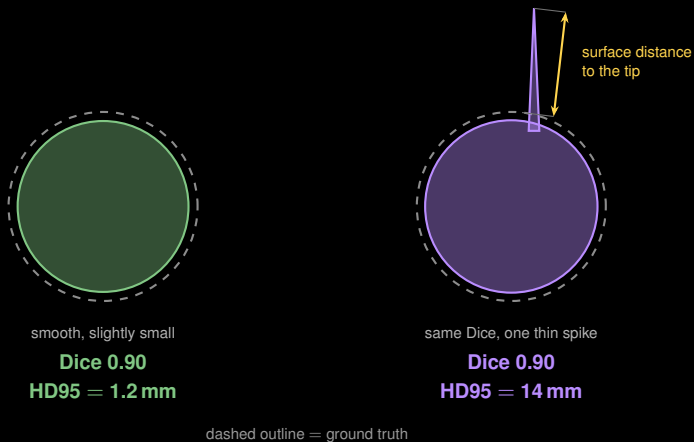


From finding **nothing** to finding the tumour, accuracy moves **0.997** to **0.999**. Dice moves **0.000** to **0.880**.

Dice and IoU are both appropriate — accuracy is not.

OVERLAP IS NOT ENOUGH

Dice and IoU measure **overlap**, **not boundary accuracy**. Two predictions, **identical Dice**:



The spike is **2.7% of the area**, so Dice barely moves — but **31% of the boundary**.

Both panels are constructed to score Dice 0.90 against the same ground truth. Point after VeraDP, *Dice Coefficient vs. IoU in Medical Image*

Segmentation, citing Kocak et al. 2025.

HAUSDORFF DISTANCE AND HD95

Distance-based measures score the **spatial distance between predicted and ground-truth boundaries**.

$$\text{HD} = \max\left(\sup_{p \in P} \inf_{g \in G} d(p, g), \sup_{g \in G} \inf_{p \in P} d(g, p)\right)$$

the maximum distance from a boundary point of one set to the closest point of the other — the **worst-case** boundary error

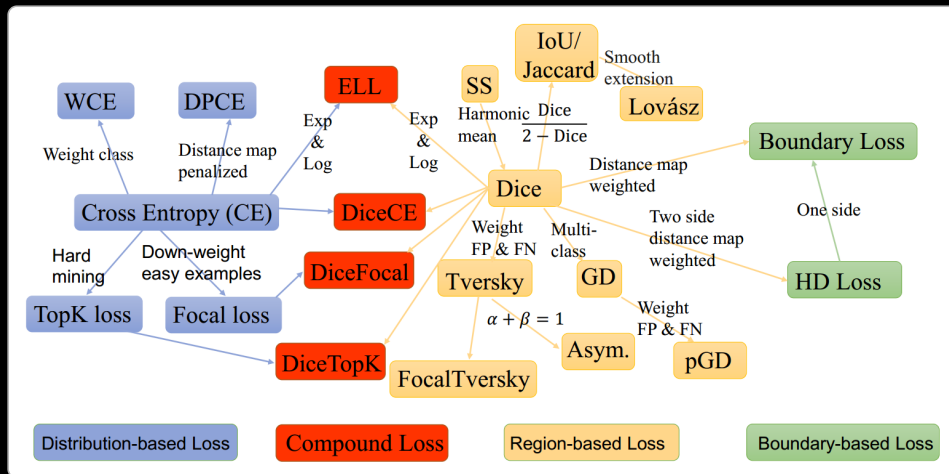
HD95 — the maximum is susceptible to outliers, so report the **95th percentile** instead.

It is measured in millimetres: resample to a common voxel spacing before comparing.

P, G are the predicted and ground-truth boundary point sets. HD: Hausdorff (1914); HD95: Taha & Hanbury (2015). HD95 is the standard boundary metric in BraTS and the Decathlon.

Training it

THE LOSS: CLASSIFY EVERY PIXEL



Distribution-based, region-based, boundary-based — and the compound losses that mix them.

Loss taxonomy: Ma, Chen, Ng, Huang, Li, Li, Yang & Martel, *Loss odyssey in medical image segmentation*, Medical Image Analysis 71 (2021)

102035; figure from the authors' repository, github.com/JunMa11/SegLoss.

CROSS-ENTROPY VS. DICE LOSS

Pixelwise cross-entropy

Treats every pixel as an independent classification.

Good: smooth, well-behaved gradients everywhere.

Bad: a tiny lesion is drowned by background, and the cheapest way to lower the loss is to predict background everywhere.

Dice loss

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_i p_i g_i + \epsilon}{\sum_i p_i + \sum_i g_i + \epsilon}$$

Optimises the metric directly, using soft probabilities p_i so it stays differentiable.

Good: scale-invariant, so a small structure still matters.

Bad: noisy gradients when the structure is absent or tiny.

The ϵ is not cosmetic — it is what keeps an empty ground-truth slice from dividing by zero.

IN PRACTICE: USE BOTH

The default

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Dice}}$$

CE supplies dense, stable gradients early in training; Dice stops the model from collapsing to background. This is what nnU-Net uses, and it is a genuinely hard baseline to beat.

If it still collapses

Class weights in the CE term, or **focal loss** to down-weight the easy background pixels.

If boundaries leak

Add a boundary-aware term. But first check it is not simply a **resolution** problem — often the mask is fine and the output stride is too coarse.

Track **validation Dice**, not validation loss. They are not the same curve, and only one of them is the thing you care about.

TRAINING LOOP, CONCRETELY

The standard training loop; only the model, loss and metric change:

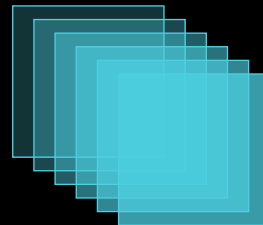
```
for x, mask in loader:
    logits = unet(x)
    loss = bce(logits, mask) \
        + dice_loss(logits, mask)
    optimizer.zero_grad()
    loss.backward()
    optimizer.step()
# predict: sigmoid(logits) > 0.5
```

- ▶ Output is a **per-pixel logit map**.
- ▶ Augment **image and mask together** — a flip that misses the mask silently corrupts the label.
- ▶ Heavy augmentation: flips, rotations, scaling, elastic deformation. Masks are scarce.
- ▶ Validate on **Dice**, and keep the best-Dice checkpoint.

Going 3D

MEDICAL IMAGES ARE VOLUMES, NOT PICTURES

- ▶ A chest CT is $512 \times 512 \times 300$. An MRI study is several such volumes, one per sequence.
- ▶ Slice it into 2D and you throw away **through-plane context** — but a lesion is a 3D object, and radiologists scroll for a reason.
- ▶ A structure that looks like a nodule on one slice is often obviously a vessel two slices away.

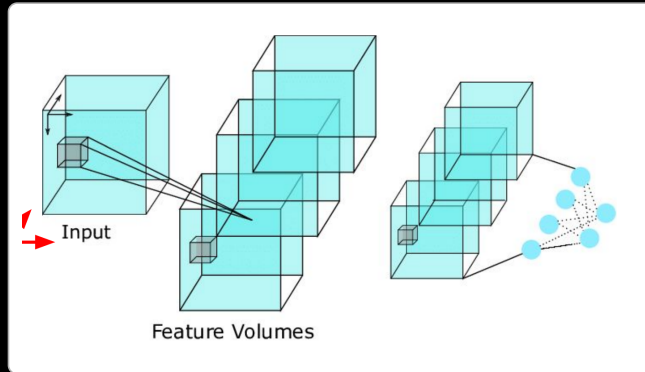


one volume,
hundreds of slices

And one more thing from Module 2

Those 300 slices are **one** training example, not 300. Nothing about going 3D changes how little data you actually have.

FROM 2D TO 3D CONVOLUTION



2D convolution

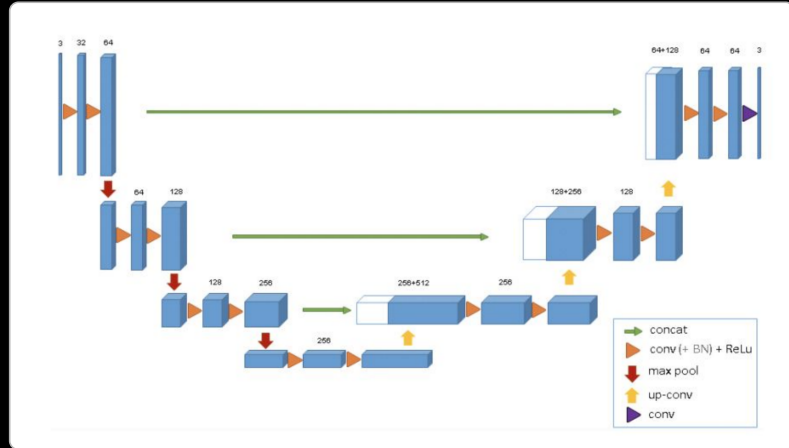
A $k \times k$ filter slides over height and width, and spans the *full* channel depth. Two directions of movement.

3D convolution

A $k \times k \times k$ filter slides along x , y **and** z . Three directions — depth is now a spatial axis, not a channel axis.

3D U-NET

Same structure as the 2D version — every 2D operation replaced by its 3D counterpart.



What changed

conv $3^2 \rightarrow 3^3$, max pool $2^2 \rightarrow 2^3$, up-conv $2^2 \rightarrow 2^3$. The encoder-decoder shape and the skip connections are untouched.

What did not

The idea. Everything you learned about the 2D U-Net transfers directly — which is rather the point.

TRAINING FROM SPARSE ANNOTATION

The real obstacle

Annotating a volume voxel-by-voxel is brutal. Nobody is going to contour 300 slices per patient for your training set.

The trick

Annotate a **few slices** per volume. Weight the loss so unlabelled voxels contribute **zero** — the network still learns 3D filters, from 2D labels.

The headline number

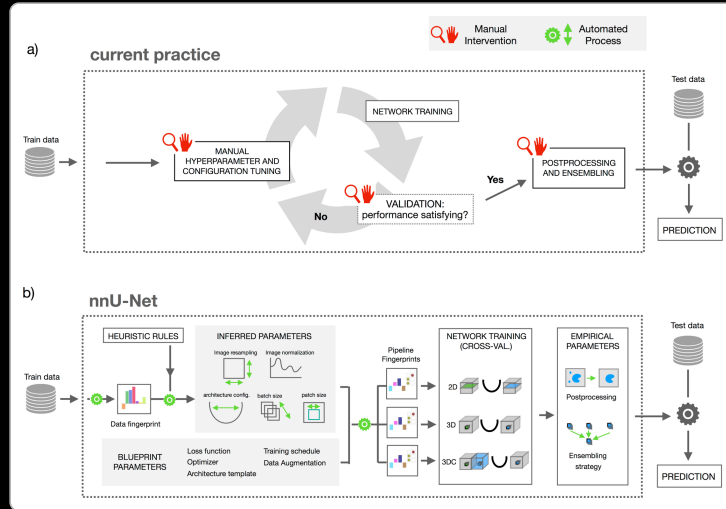
Xenopus kidney segmentation, from **3 volumes** with a total of **77 annotated slices**.

It works because one volume contains many instances of the same repeating structure — so a few slices already show the network most of the variation it needs.

This is the scarce-label problem from Module 2, answered a fourth way: **label less of each example**.

**nnU-Net: stop designing
architectures**

nnU-Net: CONFIGURE THE PIPELINE, NOT THE NETWORK



(a) current practice

A manual trial-and-error loop: tune, train, look at validation, tune again. The red hands are the human.

(b) nnU-Net

Fingerprint the data, run the rules, train 2D / 3D / cascade, pick the ensemble. **No hands**. It uses a plain U-Net — hence *no new net*.

WHAT nnU-NET ACTUALLY DECIDES

Fixed	same for every dataset	plain U-Net, loss = CE + Dice, SGD with Nesterov momentum, deep supervision (a loss at every decoder level), the augmentation set
Rule-based	computed from the fingerprint	patch size, batch size, network depth and downsampling, target spacing and resampling, intensity normalisation
Empirical	chosen by cross-validation	2D vs 3D vs 3D-cascade, which models to ensemble, whether to apply connected-component postprocessing

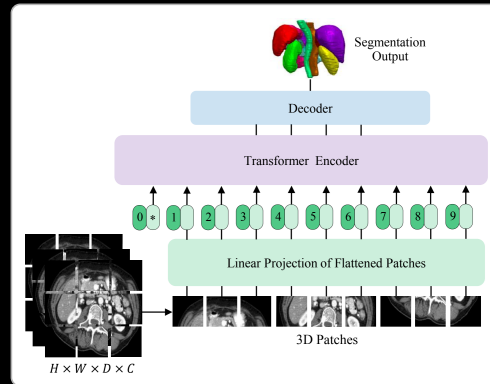
TRAINED FROM SCRATCH ON CHALLENGE DATA ALONE

33/53 new state of the art **23** datasets **11** challenges

Elsewhere, on par with or close to the top entry — organs, tumours and cells, in MRI, CT, electron and fluorescence microscopy.

Other transformers in medical image segmentation

UNETR: A TRANSFORMER ENCODER

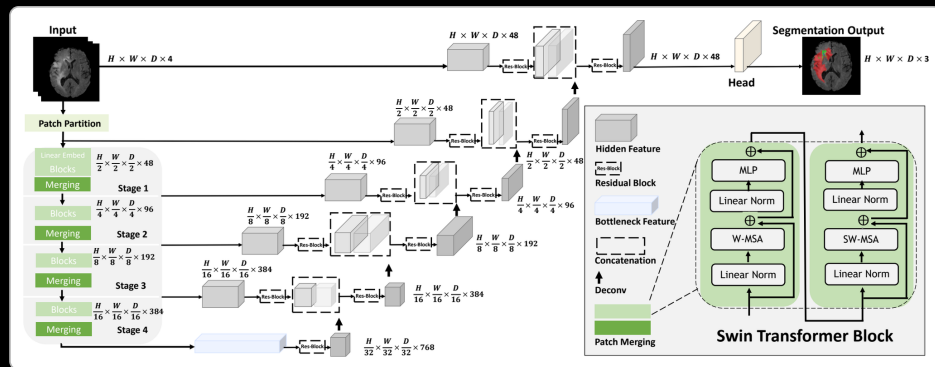


- ▶ Convolution is **local**.
- ▶ The volume becomes **3D patches**.
- ▶ A **transformer** encodes them.
- ▶ A CNN decoder rebuilds the mask.

Only the encoder changed.

Figure 1, Hatamizadeh et al., *UNETR: Transformers for 3D Medical Image Segmentation*, WACV 2022 (arXiv:2103.10504).

SWIN UNETR: ATTENTION IN SHIFTED WINDOWS

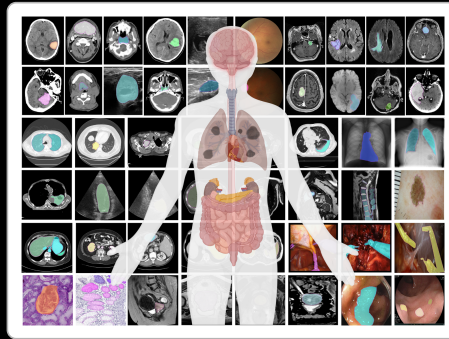


Attention inside **shifted windows**. Four stages, five resolutions, same CNN decoder.

Built for BraTS, where tumours vary in size.

Figure 1, Hatamizadeh et al., *Swin UNETR: Swin Transformers for Semantic Segmentation of Brain Tumors in MRI Images*, BrainLes/MICCAI 2021 (arXiv:2201.01266).

OR DO NOT TRAIN AT ALL: MEDSAM



- ▶ One model, any modality.
- ▶ Prompt it with a **bounding box**.
- ▶ **No training.**

1.57M image–mask pairs

10 modalities

30+ cancer types

Expert annotation time — **82% less**.

Figure 1, Ma et al., *Segment anything in medical images*, Nature Communications 15, 654 (2024), CC BY 4.0.